

Bart Gajderowicz, PhD

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Address: Toronto, Ontario

Citizenship: Canadian, Polish

PROFESSIONAL SUMMARY

I have advanced experience in software development and knowledge modelling, focusing on simulation, reasoning, machine learning, data analytics, and system architectures. I am highly skilled at problem-solving with analytical reasoning. I enjoy teaching through student engagement and experiential learning methodologies. I have extensive experience in outreach, collaboration, and bridging the gap across different fields of study and application domains. My research focuses on modelling processes, organizational behavior, and computational sociology, with applications in smart cities and social service provisioning.

RESEARCH INTERESTS

Artificial Intelligence, distributed systems, distributed problem decomposition, event-driven microservices, organizational behavior, knowledge representation (symbolic), machine learning (sub-symbolic), hybrid-AI (neural-symbolic), computational sociology, social simulation, AI planning, multi-agent systems, ontology engineering, knowledge graphs, software engineering, human decision-making augmentation, cognitive science, and human-system interactions in complex system-of-systems paradigm.

EDUCATION

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|------|---|
| 2019 | Doctor of Philosophy in Industrial Engineering
University of Toronto
Thesis: Artificial Intelligence Planning Techniques for Emulating Agents with Application in Social Services
Supervisors: Mark S Fox, Michael Gruninger |
| 2011 | Master of Science in Computer Science
Toronto Metropolitan University (formerly Ryerson University)
Thesis: Using Decision Trees for Inductively Driven Semantic Integration and Ontology Matching
Supervisors: Alireza Sadeghian, Mikhail Soutchanski |
| 2008 | Bachelor of Science in Computer Science
Toronto Metropolitan University (formerly Ryerson University) |

RESEARCH FUNDING

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| 2024 - 2026 | Public Health Agency of Canada (PHAC) (3 years; of a \$3mm project) (\$55,000/y CAD) |
| 2021 - 2022 | Mitacs Accelerate Postdoctoral Fellowship Award (1 year) (\$25,000/y CAD) |
| 2019 - 2021 | Mitacs Accelerate Postdoctoral Fellowship Award (2 years) (\$25,000/y CAD) |
| 2019 - 2021 | Wondeur AI, Postdoctoral Fellowship Award (2 years) (\$75,000/y CAD) |

AWARDS / GRANTS / ACHIEVEMENTS

2025	Recognized Reviewer Award: Text2Story 2025 Workshop
2020	Winner: Best Poster: Post-Doctoral Fellow Poster Competition Natural Sciences and Engineering Category, Lakehead University
2019	MIE Postdoctoral Fellowship Award (4 months) (\$23,000 CAD)
2017 - 2018	ACM SIGSIM PADS Travel Grant Award (\$1,000 USD)
2017 - 2018	MIE Doctoral Completion Award (\$9,500 CAD)
2013 - 2018	Research Assistantship, U of Toronto, Industrial Engineering (\$13,000/y CAD)
2013 - 2017	U of Toronto Fellowship, Mechanical and Industrial Engineering (\$9,500/y CAD)
2017	Travel Grant, AAAI Conference 2017 (\$450 CAD)
2010 - 2011	Ontario Centres of Excellence First Job Initiative (\$37,500 CAD)
2008 - 2010	Graduate Stipend, Ryerson University, Computer Science (\$12,000/y CAD)
2009	Travel Grant, International Semantic Web Conference 2009 (\$1,155 USD)
2009	Travel Grant, GECCO 2009 (\$750 CAD)
2008	Apple Worldwide Developer's Conference Student Scholarship (\$1,633 CAD)
2004	TD Bank Continuing Education Assistance Award (\$1,314 CAD)
2002	TD Bank Continuing Education Assistance Award (\$1,797 CAD)

PROFESSIONAL RESEARCH EXPERIENCE

August 2025	Assistant Professor (Temporary) , School of Computing, Utah State University
August 2025	Adjunct Assistant Professor (Status only) , Mechanical and Industrial Engineering, University of Toronto.
2023 - 2025	Senior Research Associate , Industrial Engineering Department, University of Toronto I lead the development of a hybrid AI architecture for distributed decision support and natural language understanding, focusing on extracting structured information from unstructured text using the blackboard architecture, knowledge graphs, and NLP-to-SPARQL interfaces. My technical work also includes designing data translation pipelines for integrating diverse datasets into unified knowledge graph representations, alongside mentoring students and contributing to ontology standards and technical strategy.
2021 - 2023	Postdoctoral Fellow , Industrial Engineering Department, University of Toronto This research focused on integrating applied ontologies with machine learning to automate the generation of individualized service plans by extracting entities and semantic roles from unstructured text and mapping them to a knowledge graph. The work involved building ETL pipelines, designing a hybrid symbolic/sub-symbolic model for ontology-driven knowledge extraction, and developing a methodology for dynamic needs-based planning.

- 2019 - 2021 **Postdoctoral Fellow**, Computer Science, Lakehead University, Wondour AI
 This research advanced the integration of machine learning, natural language processing, social simulation, and game-theoretical modeling to characterize market behaviors driven by subjective metrics, social norms, and emergent trends inadequately captured by traditional data sources. A simulation framework incorporating social indicators was developed to support the analysis of complex socio-economic systems, enabling robust, data-informed modeling and interpretation for effective knowledge dissemination to academic and industry stakeholders.
- 2019 **Postdoctoral Fellow**, Mechanical and Industrial Engineering,
 University of Toronto and Tata Consultancy Service (TCS), Behavioural Business
 and Social Sciences
 This research investigated hybrid computational models of decision-making in socioeconomically constrained urban contexts, emphasizing fine-grained, goal-directed agent behaviors informed by behavioral science. The work advances agent modeling by integrating planning algorithms such as heuristic planning, case-based reasoning, goal reasoning, just-in-time hierarchical planning, and the wisdom hierarchy to overcome limitations of myopic and resolute strategies.
- 2013 - 2019 **Research Assistant**, Industrial Engineering, University of Toronto
 This research focused on the development of simulation-based methodologies for analyzing and evaluating social service intervention programs, integrating interdisciplinary insights and data-driven modeling techniques. Responsibilities included defining technical objectives, identifying solution frameworks, facilitating data partnerships, and contributing to methodological advancements in program evaluation and process analysis.
- 2007 - 2010 **Senior Research Associate**, Computer Science Department,
 Toronto Metropolitan University
 This work involved the design and implementation of advanced computational systems, including intelligent matching algorithms, context-aware content management platforms, remote robotic control interfaces, and distributed media capture frameworks. Responsibilities encompassed technical leadership, system architecture, mentorship in research development, and dissemination of results to academic and industry stakeholders.

PUBLICATIONS

1. Fox, M. S; **Gajderowicz, B.**; and Lyu, D., A Maturity Model for Urban Dataset Metadata. *International Journal of Digital Curation*, 19(1): 1–27. 2025.
2. **Gajderowicz, B.**; Fox, M. S; and Gao, Y. Towards an Ontology of Traceable Impact Management in the Food Supply Chain. In *Integrated Food Ontology Workshop (IFOW 2025), part of the Proceedings of the Joint Ontology Workshops (JOWO) - Episode XI, part of the Proceedings of the Joint Ontology Workshops (JOWO) - Episode XI, co-located with the 15th International Conference on Formal Ontology in Information Systems (FOIS 2025), September 8-9, Catania, Italy*, pages 1–14, 2025.
3. **Gajderowicz, B.**; Bhardwaj, A.; and Fox, M. S RAG and Ontologies for Information Retrieval: A Literature Review. In *2nd Annual Workshop on LLMs and Ontologies, part of the Proceedings of the Joint Ontology Workshops (JOWO) - Episode XI, co-located with the 15th International Conference on Formal Ontology in Information Systems (FOIS 2025), September 8-9, Catania, Italy*, pages 1–14, 2025.

4. Knowles, P., **Gajderowicz, B.** & Dugas, K. (2024) "Data-Centric Design: Introducing an Informatics Domain Model and Core Data Ontology for Computational Systems," In *Proceedings of the 10th International Conference on Data Mining & Knowledge Management* (Vol. 14, No. 17), Conference Proceedings.
5. Rosu, D., **Gajderowicz, B.**, & Fox M.S. (2023). Representing Goals, Needs and Outcomes in Social Work. In *F. Toyoshima, M. Katsumi, E. Sanfilippo (Eds.), The Ninth Joint Ontology Workshops (JOWO 23)*, CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3637/paper15.pdf>
6. Fisher, A., **Gajderowicz, B.**, Latimer, E., Aubry, T., & Mago, V. (2022). BEAUT: An Explainable Deep Learning Model for Agent-Based Populations With Poor Data, *Knowledge-Based Systems*, 248. <https://doi.org/10.1016/j.knsys.2022.108836>
7. **Gajderowicz, B.**, Rosu, D., & Fox, M. S. (2022). Compass Event, Client, and Service Ontology: A Design Pattern for Social Services. In *T. P. Sales & M. M. Hedblom (Eds.), The Eighth Joint Ontology Workshops (JOWO 22)*, CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3249/paper2-OSS.pdf>
8. Rosu, D., Fox, M. S., & **Gajderowicz, B.** (2022). Compass Needs Ontology: A Design Pattern for Representing Needs in Social Work. In *T. P. Sales & M. M. Hedblom (Eds.), The Eighth Joint Ontology Workshops (JOWO 22)*, CEUR Workshop Proceedings. <https://ceur-ws.org/Vol-3249/paper1-OSS.pdf>
9. Fox, M. S., **Gajderowicz, B.**, Rosu, D., Turner, A., Lyu, L., & Lyu, D. (2022). An Ontological Approach to Analysing Social Service Provisioning, In *2022 IEEE International Smart Cities Conference (ISC2)*, 1-7. <https://doi.org/10.1109/ISC255366.2022.9922132>
10. Fox, M.S., Chowdhury, A., Zhang, J., **Gajderowicz, B.**, Abdulai, T., and Rosu, D. (2020). CAFO: The Common Approach Foundation Ontology, *Technical Report, Centre for Social Services Engineering, University of Toronto*. Available at csse.utoronto.ca
11. Fox, M.S., Chowdhury, A., Zhang, J., **Gajderowicz, B.**, Abdulai, T., Ruff, K., and Rosu, D. (2020). CACO: The Common Approach Core Ontology for Modeling Impact Models, *Technical Report, Centre for Social Services Engineering, University of Toronto*. Available at csse.utoronto.ca.
12. Fox, M.S., **Gajderowicz, B.**, and Ruff, R. (2020). Common Approach Indicator Vocabulary, *Technical Report, Centre for Social Services Engineering, University of Toronto*. Available at csse.utoronto.ca.
13. **Gajderowicz, B.** (2019). Artificial Intelligence Planning Techniques for Emulating Agents with Application in Social Services (Doctoral dissertation). Available at <http://hdl.handle.net/1807/97017>
14. **Gajderowicz, B.**, Fox, M.S., & Grüninger, M. (2018). The role of goal ranking and mood-based utility in dynamic replanning strategies. *Journal of Advances in Cognitive Systems*, (8), 211-230. Available at <http://cogsys.org/journal/volume6/article-6-14.pdf>
15. **Gajderowicz, B.**, Fox, M. S., & Grüninger, M. (2018). Ontology of social service needs: Perspective of a cognitive agent. In *Proceedings of the 2018 Joint Ontology Workshops, Cognition And Ontologies + Explainable AI* (pp. 1-12). Cape Town. Available at http://ceur-ws.org/Vol-2205/paper13_caos2.pdf
16. **Gajderowicz, B.**, Fox, M. S., & Grüninger, M. (2017). General model of human motivation and goal ranking. In *Proceedings of the 2017 AAAI Fall Symposium Series on a Standard Model of the Mind*, Arlington, VA: AAAI Press. Available at <https://cdn.aaai.org/ocs/15992/15992-69906-1-PB.pdf>
17. **Gajderowicz, B.**, Fox, M. S., & Grüninger, M. (2017). Requirements for emulating homeless client behaviour. In *the Proceedings of the AAAI Workshop on Artificial Intelligence for Operations Research and Social Good* (p. 7). San Francisco, CA: AAAI Press. Available at <http://eil.mie.utoronto.ca/wp-content/uploads/2015/06/BartG-socialgood-ws.pdf>

18. **Gajderowicz, B.**, Fox, M. S., Grüninger, M. (2014). Requirements for an Ontological Foundation for Modelling Social Service Chains. In *the Proceedings of the 2014 Industrial and Systems Engineering Research Conference*, Y. Guan, J. Liao (eds.), Montréal, Quebec. Available at <https://www.proquest.com/docview/1622308598>
19. **Gajderowicz, B.** (2011). Using decision trees for inductively driven semantic integration and ontology matching, (Master's thesis), Ryerson University, 350 Victoria Street, Toronto, Ontario, Canada. Available at https://rshare.library.torontomu.ca/articles/thesis/Using_decision_trees_for_inductively_driven_semantic_integration_and_ontology_matching/14656047
20. **Gajderowicz, B.**, Sadeghian, A. (2009). Ontology Granulation Through Inductive Decision Trees. In *Proceedings of the 4th International Semantic Web Conference Workshop on Uncertainty Reasoning for the Semantic Web*, Washington D.C, USA, pp. 39-50. Available at <https://ceur-ws.org/Vol-527/paper4.pdf>
21. **Gajderowicz, B.**, Sadeghian, A., Soutchanski, M. (2008). Ontology Enhancement Through Inductive Decision Trees. In *Uncertainty and Reasoning for the Semantic Web II*, da Costa, P.C.G., d'Amato, C., Fanizzi, N., Laskey, K.B., Laskey, K.J., Lukasiewicz, T., Nickles, M., Pool, M. (eds.), ISWC International Workshop, URSW 2008-2010 Revised Selected and Invited Papers. LNCS (LNAI). Springer, Heidelberg. Available at https://link.springer.com/chapter/10.1007/978-3-642-35975-0_14
22. Rahnama, H., Madni, A.M., Sadeghian, A., Mawson, C., **Gajderowicz, B.** (2008). Adaptive context for generic pattern matching in ad hoc social networks. In *Proceedings of the 2008 IEEE 3rd International Symposium on Communications, Control and Signal Processing, ISCCSP 2008*, art. no. 4537195, pp. 73-78 (2008). Available at https://link.springer.com/chapter/10.1007/978-3-642-35975-0_14

UNDER REVIEW / IN PROGRESS

1. Gajderowicz, B., Knowles, P. and Dugas, K., "Core Data Ontology for a decentralized, secure, and auditable data transformation and alignment architecture." (in progress).
2. **Gajderowicz, B.** Fisher, A. and Mago, V., "Graph pruning for identifying COVID-19 misinformation dissemination patterns and indicators on Twitter." (in progress)
3. **Gajderowicz, B.**, "Enhancing Social Work with Evidence-driven Artificial Intelligence: Literature Review" (in progress)
4. **Gajderowicz, B.**, Rosu, D., and Fox, M.S., "COMPASS Ontology: A Common Language for Collaboration and Data Sharing in Client-Centered, Outcome-focused Service Provisioning" (in progress)
5. **Gajderowicz B.** and Barton, A., "General Service Ontology Pattern" (in progress).

POSTERS

1. Mark Fox, **Bart Gajderowicz**, and Dishu Lyu: "Canadian Urban Data Catalogue," *International Association for Social Science Information Service and Technology (IASSIT)*, May 30, 2024.
2. **Gajderowicz, B.**, Fox, M.S., and Grüninger, M. (2018). "Limitations of human-centric decision making: An observer's perspective," In *Poster Proceedings of the 6th Conference on Advances in Cognitive Systems* (pp. 1-12). Stanford, CA: Cognitive Systems Foundation. Available at <http://www.cogsys.org/posters/2018/poster-2018-7.pdf>
3. **Gajderowicz, B.**, Sadeghian, A., and dos Santos, M. (2009). Expectation Maximization Enhancement with Evolution Strategy for Stochastic Ontology Mapping. In *Proceedings of the 11th Annual*

Conference on Genetic and Evolutionary Computation. ACM, New York, NY, pp. 1847-1848.
Available at <https://dl.acm.org/doi/abs/10.1145/1569901.1570197>

PATENTS AND STANDARDS

1. Fox, M.S., **Gajderowicz, B.**, and Lyu, Dishu, “Metadata Requirements for Urban Datasets”, In *Digital Governance Standards Institute Technical Committee 6 (TC 6) on Connected Cities*, CAN/DGSI 121:202X, 2025
2. **Gajderowicz, B.**, and Vijay, M. Intent elicitation in dynamic and heterogeneous networks with imperfect information. November 10 2022. US Patent App. 17/734,929 (pending), Available at <https://patents.google.com/patent/US20220358602A1/en>

INVITED TALKS AND PANELS

1. **Speaker:** Myles Harris, **Bart Gajderowicz**, David Hostler, Alexander Konovalov, “Applying RAG & Knowledge Graphs into Enterprise”, Invited workshop, May 27, 2025.
2. **Speaker: Bart Gajderowicz**, Daniela Rosu, Mark S Fox: “Enhancing Social Work with Evidence-driven Artificial Intelligence: Benefits and Concerns,” *Evidence into Practice Special Interest Group (EIPSIG)*, as part of *European Social Work Research Association (ESWRA)*, October 29, 2024.
3. **Co-Author:** Daniela Rosu, **Bart Gajderowicz**, Mark S Fox: “Empowering Social Workers: Harnessing Knowledge and Data to Improve Client Outcomes,” The International Journal on Homelessness Conference, Santiago, Chile, January 20, 2025
4. **Speaker: Bart Gajderowicz**, Mark S Fox, and Yongchao Gao: “Towards an Ontology of Traceable Impact Management in the Agriculture Food Chain,” *Foundations for Food and Food Systems Ontologies Workshop*, May 21, 2024.
5. **Speaker: Bart Gajderowicz**, Mark S. Fox, and Dishu Lyu: “Dataset Metadata Maturity Model”, *International Association for Social Science Information Service and Technology (IASSIT)*, May 31, 2024.
6. **Speaker: Bart Gajderowicz** and Mark S. Fox: “SMILE: An Explainable Artificial Intelligence System for Extracting Social Service Impact Models from Text,” *13th European Conference for Social Work Research*, April 19, 2024.
7. **Penalist:** “Integrative knowledge, ontology, and data for ecosystem and policy that ensure lifelong health, wealth and wellbeing and multi-level resilience,” *Ecosystem and Policy: Faculty of Management, Precision Convergence-Retailing*, McGill University. Panelists: Andrea Borghini, **Bart Gajderowicz**, Damien Dooley, John Keogh, February 2024.
8. **Speaker: Bart Gajderowicz:** “Towards an Ontology of Traceable Impact Management in the Agriculture Food Chain.” *Series: The Convergent Innovation Webinar Series: Inventing “One-World” Solutions for Lifelong Wellness and Sustainable Economic Growth*. Panelists: Jurij Skornik, Troy Ruemping, Rob Warren, Horia Bradau, Naomi Kay Fukagawa. December 6, 2023
9. **Panelist:** Andrea Borghini (speaker): “Bots in the Kitchen: Philosophical Take on the Digital Food Transformation.” *Series: The Convergent Innovation Webinar Series: Inventing “One-World” Solutions for Lifelong Wellness and Sustainable Economic Growth*. Panelists: Brian King, Catherine Paquet, **Bart Gajderowicz**, Ebenezer Miezah Kwofie. September 23, 2024.
10. **Speaker: Bart Gajderowicz.** “Survey of University of Toronto Ontology of Impact,” *Government Ontology workshop, Basic Formal Ontology Summit Meeting*. May 23, 2023

11. **Speaker:** Matt Parker, **Bart Gajderowicz**, Lester Lym, Daniela Rosu, Alina Turner & Mark S. Fox ,
“Data Consolidation: How we Solved it with Kompas Ontology and Data,” *6th Annual Data That Makes a Difference Conference*, Calgary, Alberta, Canada, May 17th, 2023
12. **Seminar:** **Bart Gajderowicz**. ”Neural-symbolic artificial intelligence: state-of-the-art, what’s missing, and next steps,” *Department of Computer Science Graduate Seminar 2022*, Lakehead University, Thunder Bay, Ontario, Canada, September 23, 2022.
13. **Panelist:** “Research that is outside of the box,” *Research and Innovation Week*. Panelists: Salimur Choudhury, Martha Dowsley, **Bart Gajderowicz**, Lindsay Galway, Lana Ray, Pauline Sameshima, Lakehead University, Thunder Bay, Ontario, Canada. February 28, 2020
14. **Speaker:** **Bart Gajderowicz**, Mark S. Fox, and Michael Grüninger. *Artificial Intelligence Planning Techniques for Emulating Agents*: ACM SIGSIM PhD Colloquium, Chicago, Illinois, USA, June 4, 2019 (Awarded an ACM Travel Grant)
15. **Speaker:** **Bart Gajderowicz**, Mark S. Fox, and Michael Grüninger. *Requirements for Data-Driven Social Service Policy Evaluation*. Third Annual Data Sharing Initiative Calgary, Alberta, Canada, May 28th, 2018
16. **Panelist:** *Ethics in Artificial Intelligence*. Panelists: Dwija Patel, Matt Kantor, **Bart Gajderowicz**, and Myles Harrison. Toronto, Ontario, Canada, July 13, 2017
17. **Panelist:** *The Future of Semantic Web and its Applications*, Panelists: Mark van Berkel, **Bart Gajderowicz**, and James Leigh; Toronto, Ontario, Canada, April 4, 2017
18. **Speaker:** **Bart Gajderowicz**, Mark S. Fox, and Michael Grüninger. *Artificial Intelligence Planning Techniques for Emulating Agents, with Application in Social Services*. Machine Intelligence Toronto. Toronto, Ontario, Canada, March 9, 2017.

SERVICE AND COMMITTEES

2023 - 2025	Technical Committee: Common Approach: Common Impact Data Standard
2022 - 2025	Workshop Chair: Workshop on Ontologies for Services and Society (OSS)
2023 - 2025	Program Committee: Formal Ontology in Information Systems (FOIS)
2024 - 2025	Program Committee: Applied Ontology Journal
2024	Program Committee: International Conference on Biological and Biomedical Ontology
2024	Program Committee: International Association for Social Science Information Service and Technology (IASSIST) Quarterly
2023 - 2024	Program Committee: International Workshop on Narrative Extraction from Texts
2022	Program Committee: Semantic Web Journal
2019	Steering Committee: EAI International Conference on Social Data and AI
2015 - 2016	Steering Committee: MIE Graduate Student Research Symposium
2019 - 2021	Review Committee: Computational and Mathematical Organization Theory
2015 - 2020	Steering Committee: Undergraduate Engineering Research Day, U of Toronto

MENTORING AND SUPERVISION ACTIVITIES

I have mentored and assisted in the supervision of PhD, Master's, and undergraduate students on research projects, focusing on developing their research and software engineering skills. I have advised on publications strategies and venues, and proofread numerous manuscripts, advising one student on a successful PhD grant application. Worked closely with two students on their academic and software development and mentored/supervised over 25 students on their work-study research projects.

PhD Students

- 2023 - 2024 Ranfateh Chattha: PhD Student, Political Science, University of Toronto. Supervised on development of annotation vocabulary for social service and machine learning data corpus (2024).
- 2023 - 2024 Liwei Liao: PhD Student, Educational Leadership and Policy, University of Toronto. As a work-study position, I supervised Liwei on data annotation for an NLP project. Liwei was exposed to the CIDS impact model ontology, which was used to annotate unstructured text for testing and training purposes (2023-2024).
- 2023 - 2025 Yunhong Tian: PhD Student, Civil Engineering, University of Toronto. As a work-study position, I supervised Yunhong as an Urban Data Curation, metadata and taxonomy creation, geospatial and social service dataset analysis.
- 2020 Gaurav Rao: PhD Student, Applied Sciences, Saint Mary's University. Assisted with a successful Mitacs grant application for PhD Research (2020).

Master's Students

- 2024 Jianyue Bai: MEd in Developmental Psychology and Education, University of Toronto. Supervised on urban data curation, metadata modeling and taxonomy creation.
- 2024 Hussein Ismail: Bachelor of Applied Science, Industrial Engineering, University of Toronto. Supervised on urban data curation, metadata modeling and taxonomy creation.
- 2022 - 2023 Girirah Heda: MSc Student in Master's of Information, School of Information, University of Toronto. Supervised on urban data curation, metadata modeling, taxonomy creation, data analysis, and software engineering.
- 2021 Marina Silic: MSc Student, Medical Biophysics The University of Toronto. As a work-study position, I supervised Marian on hypothesis testing and ETL pipeline on the hybrid NLP project.
- 2019 Pedram Khoshnevis: assisted in the supervision of Computer Science Master's Thesis, Lakehead University. Defended in April 2021.
- 2019 Dhivya Chandrasekaran: MSc Student, Computer Science, Lakehead University. Assisted with editing numerous manuscripts. Currently a PhD student in Computer Science, Saint Mary's University.
- 2019 - 2020 Andrew Fisher: MSc Student, Computer Science, Lakehead University. Assisted in the supervision of. Master's thesis. Defended in December, 2020. In 2024, graduated with a PhD from the Computer Science Department, Saint Mary's University. Currently a Postdoctoral Fellow at York University, Toronto, Canada.

Undergraduate Students

- 2024 - 2025 Naoroj Farhan: Bachelor of Science, Computer Science, Mathematics, University of Toronto. Supervised on experiments for large language model training and testing for entity recognition analysis and system mapping data for AI training and testing corpus.
- 2024 – 2025 Jiaming Lu, Honours BSc in Computer Science, University of Toronto. Experiment with retrieval-augmented generation (RAG) techniques for named entity recognition and SPARQL learning.
- 2024 - 2025 Chloe Tam: Honours Bachelor of Arts. Public Policy (Major), Economics and Political Science (Minors), University of Toronto. Supervised on analysis and system mapping data for AI training and testing corpus.
- 2024 - 2025 Kasandra Tworzyanski: Physical & Environmental Geography with minors in GIS and Statistics, University of Toronto. Supervised on urban data curation, metadata modeling and taxonomy creation, as well as analysis and system mapping data for AI training and testing corpus.
- 2024 – 2025 Bairavie Piravakaran Honours Bachelor of Science (HBSc), Population Health. Supervised on urban data curation, metadata modeling and taxonomy creation, as well as analysis and system mapping data for AI training and testing corpus.
- 2024 - 2025 Lian Hosseinnia: Bsc Student, global Health and Bioethics, University of Toronto. Supervised on development of annotation vocabulary for social service and machine learning data corpus, and reports for social media.
- 2023 - 2025 Yifan Liu: BSc Student, Computer Science, University of Toronto. Supervised Yifan on Natural Language Models and software engineering methodology, including unit testing and architecture design.
- 2024 Kaviraj Gaire: BSc Student, Computer Science, University of Toronto. Supervised Kaviarj during the development of a knowledge graph reasoning system for extracting impact model information from unstructured text.
- 2023 - 2024 Janel Gilani: BSc Student, Computer Science Specialist & Statistics Minor, University of Toronto. Supervised Janel on NLP Validation methods using taxonomy, graph theory, and TF-IDF encoding. Utilized GridSearch for aligning various taxonomies with
- 2023 - 2024 Silin Lyu: BSc Student, Economics and Statistics, University of Toronto. Supervised on NLP model testing, software engineering methodology and unit testing. As an urban data curator, supervised Silin on metadata and taxonomy creation, and ETL pipeline software engineering.
- 2023 - 2024 Rudraksh Monga: BSc. Student, Computer Science; Minor: Mathematics and Statistics, University of Toronto. Supervised Rudraksh on LLM training and transfer learning methods, as well as ML optimization with GridSearch.
- 2023 - 2024 Aaron Lio: BASc - Engineering Science, University of Toronto. Supervised on Prompt Engineering for Named Entity Extraction using ChaptGPT for extracting impact model concepts.
- 2023 James Hariady Widjaja: BSc Student, Computer Science & Economics, University of Toronto. Working with a linguist, I supervised James on NLP models, specifically evaluation and correction of dependency trees.
- 2023 Leonid Nediak: BSc Student, Mathematics and Computer Science Specialisation, University of Toronto. Supervised Leonid on creating a explainable NLU benchmarking

- suite, with a focus on Named Entity Extraction, with complex confusion matrix architecture and metrics.
- 2023 Sunnie Hu: BA Student, Health Studies with Statistical Studies and Visual Sciences, University of Toronto. As a work-study position, I supervised Sunnie on data annotation for an NLP project. Sunnie was exposed to the CIDS impact model ontology, which was used to annotate unstructured text for testing and training purposes.
- 2022 - 2023 Chaerin Song: BScs Student, Statistics and Linguistics, University of Toronto. Supervised Chaerin in performing hypothesis testing and developing a hybrid AI (symbolic and sub-symbolic) model for semantic relation extraction. Chaerin developed a prompt-generating process for Question/Answer LLMs.
- 2019 - 2021 Akash Shetty: BSc Student, Honors Math and Physics, McMaster University. Supervised Akash as an AI Research Intern at Wondeur AI on community detection and dimensionality reduction methods.

Professional Mentoring

- 2021 - 2024 Dishyu Lyu: Research Assistant, Urban Data Centre, School of Cities, University of Toronto. Supervise Dishyu during the developing and testing for a data integration ETL pipelines. Performed ontology engineering for semantic integration portion of the pipeline. Supervised developing and testing the Canadian Urban Data Catalogue and the Dataset Metadata Maturity Model.
- 2019 - 2021 Sharanya Ramidi: Machine Learning Engineer at Wondeur AI. Managed and mentored Sharanya on NLP tasks, creating text extraction scripts and evaluation criteria. Taught computer programming principles for data analysis.
- 2019 Suman Kumar: Software Engineer, Tata Consulting Services. As part of my postdoctoral position with TCS, I worked closely with Suman (PI) and two junior machine learning engineers: supervised and taught AI planning methods, case-based reasoning, and agent-based simulation, 2019.
- 2011 - 2014 Various employees and co-op students: As a lead semantic engineer and the CTO, I interviewed, mentored and supervised a number of software engineers.

Highschool Students

- 2008 Nilanthy Balendra, Susan Feng, and Nika Zolfaghari: While at the Computer Science Department, Toronto Metropolitan University, I mentored three high-school interns as part of the “Research Opportunity Programs in Engineering and Science and Women in Engineering” initiative.

TEACHING EXPERIENCE

Utah State University, School of Computing

- 2025 CS1410 Introduction to Computer Science--CS 2 (Java Programming)
- 2025 CS2810 Computer Systems Organization and Architecture I

University of Toronto, Mechanical and Industrial Engineering Department

- 2017 MIE457 Knowledge Modelling and Management, Teaching Assistant
- 2016 MIE1512 Data Analytics, Teaching Assistant

2016	MIE253 Data Modeling, Head Teaching Assistant
2013 - 2016	MIE350 Design and Analysis of Information Systems, Teaching Assistant
2015	MIE1513 (451) Decision Support Systems, Instructor
2014	MIE451 Decision Support Systems, Teaching Assistant

Toronto Metropolitan University, Information Technology Studies

2010 - 2011	CXCP685 PHP/MySQL Web Design Fundamentals, Instructor
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Toronto Metropolitan University, Computer Science Department

2009 - 2010	CPS125 Digital Computation and Programming, Instructor
2007 - 2010	CPS630 Advanced Web Applications, Teaching Assistant
2008	CPS109 Computer Science I, Teaching Assistant
2007	CPS209 Computer Science II, Teaching Assistant

PROFESSIONAL EXPERIENCE

2013	Senior Developer, Influitive Inc., Toronto, Ontario This work focused on system integration, scalable software design, and rapid prototyping within a production architecture, including the incorporation of A/B testing methodologies for iterative feature validation. Additional responsibilities included leading initiatives to embed machine learning into platform workflows, developing both technical frameworks and business justifications for their adoption.
2011 - 2013	Co-Founder and CTO, Engagio Inc. (acquired by Influitive Inc.) This work involved leading the end-to-end design and implementation of a high-throughput, scalable system architecture capable of processing over one million user-generated content items daily, following lean startup and agile development principles. Responsibilities included technical strategy, system performance analytics, and engineering leadership to ensure rapid iteration, scalability, and operational efficiency from prototype to production deployment.
2010 - 2012	Software Engineer, Semantic Technologies, Equentia Inc. This work involved integrating semantic technologies and natural language processing into a large-scale content processing system to enhance automated text classification and topic extraction. Responsibilities included taxonomy management, external knowledge base integration using YAGO and DBpedia, and the evaluation and deployment of academic and open-source NLP tools within a high-throughput production environment.
2001 - 2007	Solutions Developer, Enterprise Technology Solutions, TD Canada Trust Developed and maintained financial systems across the full project lifecycle using technologies such as XML, XSL, Ruby, and SAS, with responsibilities spanning project management, system analysis, and statistical data mining.

WEB APPLICATION CONSULTANT

2010	NuLayer Inc.
2009 - 2010	Celect.org
2008 - 2009	Mikemap.com - Independent Music and Event Management
2008	Athletes Video - Social Network
2007	Athletes Video - Media Management System

APPOINTMENTS

2015 - 2016	Vice President, AMIGAS, University of Toronto student group at MIE Liaison with MIE department faculty and staff, advise and coordinate Academic Committee initiatives with Academic VPs; outreach program to IE members and student groups; advisor to the MIE 2016 Graduate Research Symposium Committee.
2014 - 2015	IE Representative, AMIGAS, University of Toronto student group at MIE Representing and supporting Information Engineering (IE) initiatives.
2015	MIE 2015 Graduate Research Symposium Co-Chair.
2009 - 2010	President, Computer Science Graduate Student Association (CSGSA), TMU
2009 - 2010	CSC Graduate Student Representative, Computer Science Council, TMU

TECHNICAL SKILLS

Main Programming Languages: Python, Java, Prolog, OWL 2, RDF, Ruby, SPARQL, SQL

Main Methodologies: Ontologies (reasoning with OWL 2, RDF, SWI-Prolog, engineering with Protégé), Databases (relational DBMS, graph-based DBMS, knowledge graphs, NetworkX), simulation (agent-based modelling, GAMA), Machine Learning (BERT, Pytorch, Sci-Kit Learn), miscellaneous tools (WordNet, DBpedia, Linux Shell, Git).

PROFESSIONAL MEMBERSHIPS

2017 - present Association for the Advancement of Artificial Intelligence (AAAI)

2009 - present International Association for Ontology and its Applications (IAOA)

2009 - present Association for Computing Machinery Student Membership (ACM)

2009 - present Institute of Electrical and Electronics Engineers (IEEE)